EE 558

1. HW # 1

2. More examples.

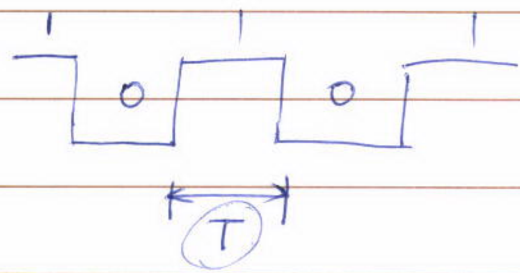
1. (a) λ: 0.4 ~ 0.7 μm

$\Rightarrow f$: Hz.

$$f_1 = \frac{c}{\lambda_1} = \frac{3e8}{0.4e-6} = 750 \text{ THz}.$$

$$f_2 = \frac{c}{\lambda_2} = \frac{3e8}{0.7e-6} = 429 \text{ THz}.$$

(b)



R_b = 1M bit/sec or 1 Gbit/sec

$\Rightarrow T$? for power level is 1mW, energy each pulse?

$T = \frac{1}{R_b}$. 1 Mb/s $\rightarrow$ $T = \frac{1}{10^6} = 1\mu s$.

1 Gb/s $\rightarrow$ $T = \frac{1}{10^9} = 1 ns$.

1 mW $\Rightarrow$ $E = T \cdot P = 1\mu s \cdot 1 mW = 1 nJ$.

$E =$ $1 ns \cdot 1 mW = 1 pJ$.

(c). glass : loss = 10^5 dB/km.

fraction of light remains after 5mm?

~~Att~~ loss = unit loss × distance.

dB? ✓

linear?

10^5 dB.

~~$\neq 10^5$ dB.~~

~~loss in dB:~~ 10^5 dB/km × 5 mm.

= 0.5 dB

0 dBm signal comes in. $\rightarrow$ 1

−0.5 dBm signal comes out $\rightarrow$ fraction.

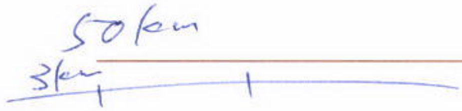
fraction $10^{\frac{-0.5}{10}} = 89\%$.

(d). $v \rightarrow n = 1.5$. $\quad v = \dfrac{c}{n} = \dfrac{3e8}{1.5}$.

2. Link budget.

Transmitted power (dBm) $\xrightarrow[\text{(dB)}]{\text{Link loss}}$ Received Power (dBm).

$\sum (+)$ $\begin{cases} \text{fiber loss (dB)} \\ \text{fusion loss/slice (dB)} \\ \text{connector losses (dB)} \end{cases}$



- Fiber loss: $50\,\text{km} \times 0.5\,\text{dB/km} = 25\,\text{dB}$.

connectors: $\left\lfloor \dfrac{50}{3} \right\rfloor \times 0.3\,\text{dB} = 4.8\,\text{dB}$.

$1 \times 2 = 2\,\text{dB}$.

Σ

Total loss: 31.8 dB.

Input power: $1\,\text{mW} \rightarrow 0\,\text{dBm}$.

Output power: $\underset{\text{power}}{0\,\text{dBm}} - \underset{\text{fraction}}{31.8\,\text{dB}} = \underset{\text{power}}{-31.8\,\text{dBm}}$.

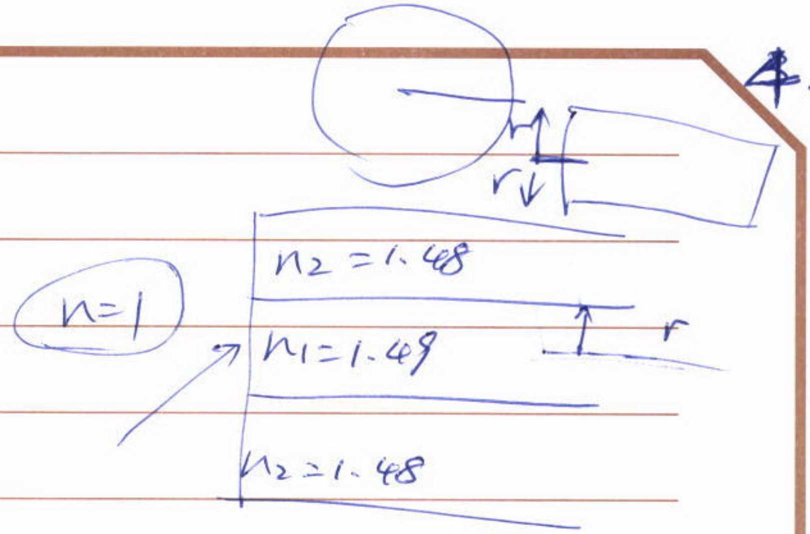
mW / O = mW

3.

$r = 100\ \mu m$.

$\lambda = 1.3\ \mu m$.

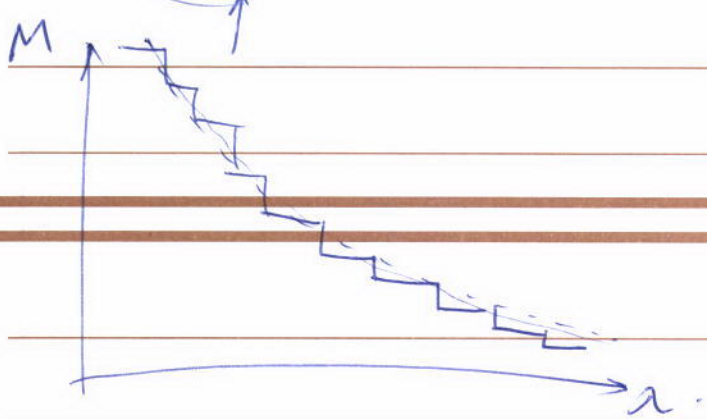
M



$V = 83.23$.

$M = \left\lfloor \frac{V^2}{4\pi^2} \right\rfloor = \underline{3k}$

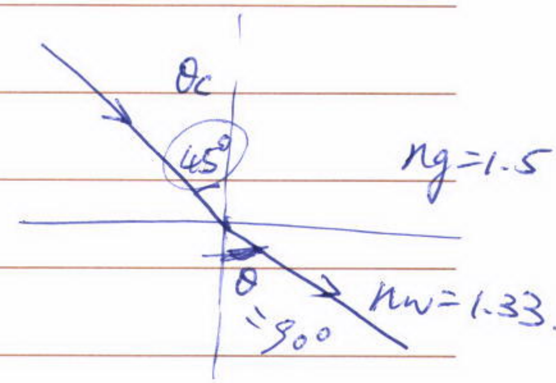
Symmetric.
step-index



Integers

sketch.

4. $\sin\theta \cdot n_w = \sin 45^\circ \cdot n_g$.



$\theta = \arcsin\left(\sin 45^\circ \cdot \frac{n_g}{n_w}\right)$

$= 52.9^\circ$.

5. $\theta_c = \arcsin\left(\sin 90^\circ \cdot \frac{n_w}{n_g}\right) \approx 62.5^\circ$.

6. (a). f ? $\Leftarrow$ 1.55 μm.

$$f = \frac{c}{\lambda} = \frac{3e8}{1.55e-6} = 193.6\,THz$$

$$\boxed{192 \sim 196\,THz}$$

(b). 0.1 nm bandwidth $\Rightarrow$ bandwidth in frequency.

$$f_1 = \frac{c}{\lambda_1} \qquad \lambda_1 \sim \lambda_2 : \underbrace{\lambda_2 - \lambda_1}_{} = 0.1\,nm.$$

$$f_2 = \frac{c}{\lambda_2} \longrightarrow f_1 - f_2 \ ? = \Delta f$$

$\frac{df}{d\lambda}$?

$d\lambda \to 0.1\,nm$.

$$df \approx f' \cdot d\lambda$$

$$= -\frac{c}{\lambda^2} \cdot d\lambda.$$

$$= -\frac{3e8}{(1.55\,\mu m)^2} \cdot 0.1\,nm.$$

$$\approx -12.5\,GHz.$$

$\Downarrow$

12.5 GHz.

1 nm	~ 125 GHz
0.8 nm	~ 100 GHz.
0.1 nm	~ 12.5 GHz

7. (a). $\lambda_1 = 0.82\mu m$, $\lambda_2 = 1.3\mu m$, $\lambda_3 = 1.55\mu m$.

↳ vacuum.

$$E = hf = \frac{hc}{\lambda} \rightarrow J \Rightarrow eV$$

1V · electron charge.

$$\begin{cases} 1eV = 1.602\times10^{-19} J \\ 1J = 6.24\times10^{18} eV \end{cases}$$

$$\lambda_1 : E_1 = \frac{h\cdot c}{\lambda_1} \cdot 6.24\times10^{18} = 1.51\ eV$$

$$\lambda_2 : E_2 = 0.955\ eV$$

$$\lambda_3 : E_3 = 0.8\ eV$$

(b) $$\lambda = \frac{hc}{E} = \frac{hc}{1.24eV \cdot 1.602\times10^{-19} J/eV} = 1.0\mu m$$

$\uparrow$ $1.24eV$

vacuum.

$$f = \frac{c}{\lambda} = 299.8\,THz$$

8. $m\lambda = 2nL$

integer vacuum refractive index cavity length.

7

3.6   430 μm.

$m\lambda = 2nL$.

(a). 1.53 μm — 1.57 μm ?

m might not be an integer

→ estimation.

m_1 : 1.53x μm. m_2 : 1.56̸9 μm

↓ as small as possible.   ↓ as large as possible.

$m_1 \left\lfloor \frac{2nL}{\lambda_1} \right\rfloor = 2023.\cancel{x}$

1.53

$m_2 \left\lceil \frac{2nL}{\lambda_2} \right\rceil = 1971.\square = 1972$

1.57

$N = m_1 - m_2 + 1 = 2023 - 1972 + 1 = 52$

(b). $|m_1 - m_2 + 1| = \underline{1} \Leftarrow L = 0.43\ \mu m.$

$m\lambda = 2 \times 3.6 \times 0.43\ \mu m.$

1.57 7.2×0.43

$m\lambda = 3.096.$

$$\begin{array}{r} 43 \\ \times\ 72 \\ \hline 86 \\ 301 \\ \hline 3096 \end{array}$$

frequency modes — Fabry-Perot cavity, laser
spatial modes — waveguide

different frequencies.

same frequency.

9. (a) $v = \frac{c}{n_1}$

Ray B: $t_B = \frac{L_B}{v} = \frac{L}{v} = \frac{L n_1}{c}$

Ray A: $t_A = \frac{L_A}{v} = \frac{L}{\sin\varphi \cdot v} = \frac{L}{\frac{n_2}{n_1} \cdot \frac{c}{n_1}} = \frac{L n_1^2}{c \cdot n_2}$

$$\Delta T = t_A - t_B = \frac{L}{\sin\varphi \cdot v} - \frac{L}{v}$$

$$= \frac{L n_1^2}{c \cdot n_2} - \frac{L n_1}{c}$$

$$= \frac{L n_1 (n_1 - n_2)}{c n_2}$$

(b). $\frac{1\,\text{km} \times 1.475\,(1.475 - 1.460)}{3e8 \cdot 1.460} = 50.55\ \text{ns}$

(c). $V = \frac{2r}{\lambda}\sqrt{n_1^2 - n_2^2}$

$M = 6 > 5$

HW2

1. Laser $L = 350\mu m$. $n = 3.4$

wavelength near $1.55\mu m$. → ossillation wavelength?

→ $m\lambda = 2nL$.

→ vacuum

① estimation.

② if m is integer → m.

else $\lfloor m \rfloor$ $\lceil m \rceil$

2. dispersions.

1° different light ~~have di~~ spend different time inside a waveguide

→ different frequency modes. < same distance L / different speed v

→ different spatial modes — Same speed v / different distance L

$D = 17\, ps/(km/nm)$

2 wavelength 1550nm. 1551 nm.

50km (1.55μm). → 1km fiber

125GHz bandwith. → $\Delta T = 17\, ps$

$\Delta T = D \cdot L \cdot BW = 17ps/km/nm \cdot 50km \cdot 1nm = 850\, ps.$